

Solutions – Tutorial: Inequality and Poverty in Canada

ECON 204 – Contemporary Economic Issues

Winter 2026

Question 1. Income Distribution and Inequality

The annual before-tax household incomes (in Canadian dollars) are:

(18,000, 22,000, 26,000, 30,000, 35,000, 42,000, 50,000, 65,000, 95,000, 180,000)

(a) Mean and Median Income

Step 1: Total income.

$$18,000 + 22,000 + 26,000 + 30,000 + 35,000 + 42,000 + 50,000 + 65,000 + 95,000 + 180,000 = 563,000$$

Step 2: Mean income.

$$\bar{y} = \frac{563,000}{10} = 56,300$$

The mean household income is **\$56,300**.

Step 3: Median income. With 10 households, the median is the average of the 5th and 6th incomes:

$$\text{Median} = \frac{35,000 + 42,000}{2} = 38,500$$

Interpretation. The median income (\$38,500) is much lower than the mean. This occurs because very high incomes at the top of the distribution pull the mean upward. For this reason, the median is often a better measure of the “typical” household income in Canada.

(b) Income Shares

Step 1: Total income.

$$Y = 563,000$$

Step 2: Bottom 40% income share (households 1–4).

$$Y_{B40} = 18,000 + 22,000 + 26,000 + 30,000 = 96,000$$

$$\text{Share}_{B40} = \frac{96,000}{563,000} \times 100 \approx 17.05\%$$

Step 3: Top 10% income share (household 10).

$$Y_{T10} = 180,000$$

$$\text{Share}_{T10} = \frac{180,000}{563,000} \times 100 \approx 31.97\%$$

Interpretation. In this economy, the top 10% of households earn nearly one-third of all income, while the bottom 40% earn only about 17%. This indicates substantial income concentration at the top.

Question 2. Simple Inequality Ratios

(a) Top-to-Bottom Ratio

$$\text{Top-to-bottom ratio} = \frac{180,000}{18,000} = 10$$

Interpretation. The richest household earns 10 times as much as the poorest household, illustrating a large income gap.

(b) Policy Change

After the policy:

$$\text{New ratio} = \frac{150,000}{25,000} = 6$$

Conclusion. The top-to-bottom ratio falls from 10 to 6, indicating that inequality decreases under the policy.

Question 3. Poverty Measurement in Canada

Poverty thresholds:

- LICO = \$26,000
- MBM = \$28,000

(a) Poverty Status

Below LICO (strictly below \$26,000):

$$18,000, 22,000 \Rightarrow 2 \text{ households}$$

Below MBM (strictly below \$28,000):

$$18,000, 22,000, 26,000 \Rightarrow 3 \text{ households}$$

(b) Poverty Rates

$$\text{LICO poverty rate} = \frac{2}{10} \times 100 = 20\%$$

$$\text{MBM poverty rate} = \frac{3}{10} \times 100 = 30\%$$

Interpretation. The MBM poverty rate is higher because the MBM threshold is higher than the LICO threshold, capturing a more demanding standard of living.

Question 4. Depth of Poverty

MBM threshold = \$28,000.

(a) Poverty Gaps

$$28,000 - 18,000 = 10,000$$

$$28,000 - 22,000 = 6,000$$

$$28,000 - 26,000 = 2,000$$

(b) Average Poverty Gap

$$\text{Average gap} = \frac{10,000 + 6,000 + 2,000}{3} = 6,000$$

Interpretation. The poverty gap measures how far households are below the poverty line, providing information about the severity of poverty, not just its incidence.

Question 5. Inflation and Poverty

(a) New MBM Threshold

$$28,000 \times 1.08 = 30,240$$

The new MBM threshold is **\$30,240**.

(b) Poverty Impact

Households below \$30,240:

$$18,000, 22,000, 26,000, 30,000 \Rightarrow 4 \text{ households}$$

$$\text{New poverty rate} = \frac{4}{10} \times 100 = 40\%$$

Interpretation. Inflation increases the cost of living and can raise poverty rates even if nominal incomes do not change.